

## QINZHOU, China

WMO: 596320

Lat: 21.9498N

Lon: 108.6138E

Elev: 20

StdP: 14.69

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 43.5       | 45.9 | 26.4                          | 20.3 | 54.9 | 30.8 | 25.0 | 53.7 | 17.6                  | 48.3 | 15.9 | 50.5 | 8.2                   | 0    | 0.323 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 9.4                    | 93.8            | 80.3 | 92.4 | 80.2 | 91.2 | 80.1 | 83.4                | 88.4 | 82.6 | 87.9 | 82.0 | 87.4 | 6.1                  | 200  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 82.1                            | 167.0 | 85.9 | 81.3 | 162.9 | 85.6 | 80.7 | 159.4 | 85.4 | 47.6          | 89.0 | 46.7 | 88.1 | 46.1 | 87.5 | 86.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |  |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|--|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |  |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |  |
| 15.9              | 13.7 | 11.8 | DB                         | 40.9 | 97.0               | 3.3 | 1.5                                                | 38.5 | 98.1       | 36.6 | 99.0       | 34.7 | 99.8       | 32.3 | 100.8 |  |
|                   |      |      | WB                         | 37.0 | 84.3               | 2.5 | 0.9                                                | 35.2 | 85.0       | 33.7 | 85.5       | 32.3 | 86.0       | 30.5 | 86.7  |  |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 73.9   | 57.9  | 60.8  | 66.3  | 74.9  | 81.0  | 83.7  | 84.3  | 84.3  | 82.6  | 78.0  | 70.5  | 61.8  |
|                                                                                  | DBStd       | 11.11  | 7.63  | 8.75  | 7.54  | 6.30  | 4.00  | 2.95  | 3.11  | 2.99  | 3.51  | 4.66  | 6.08  | 6.41  |
|                                                                                  | HDD50       | 29     | 17    | 9     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 3     |
|                                                                                  | HDD65       | 667    | 242   | 172   | 81    | 10    | 0     | 0     | 0     | 0     | 0     | 1     | 22    | 139   |
|                                                                                  | CDD50       | 8757   | 262   | 312   | 505   | 747   | 961   | 1012  | 1064  | 1063  | 977   | 869   | 615   | 370   |
|                                                                                  | CDD65       | 3921   | 23    | 55    | 121   | 308   | 496   | 562   | 599   | 598   | 527   | 404   | 188   | 40    |
|                                                                                  | CDH74       | 39142  | 25    | 152   | 467   | 2166  | 5059  | 6686  | 7293  | 7087  | 5761  | 3389  | 958   | 99    |
|                                                                                  | CDH80       | 14515  | 0     | 14    | 60    | 485   | 1768  | 2709  | 3133  | 2967  | 2211  | 1016  | 149   | 3     |
| Wind                                                                             | WSAvg       | 5.4    | 5.7   | 5.5   | 5.3   | 5.6   | 5.6   | 5.3   | 5.1   | 4.6   | 5.0   | 5.4   | 5.3   | 5.7   |
| Precipitation                                                                    | PrecAvg     | 84.1   | 1.8   | 2.1   | 2.5   | 5.7   | 8.8   | 13.5  | 16.7  | 16.4  | 8.1   | 5.0   | 2.3   | 1.2   |
|                                                                                  | PrecMax     | 110.6  | 7.9   | 9.8   | 8.2   | 15.8  | 20.7  | 33.2  | 33.4  | 37.4  | 19.1  | 15.2  | 7.6   | 4.2   |
|                                                                                  | PrecMin     | 47.4   | 0.1   | 0.1   | 0.4   | 0.6   | 2.9   | 3.7   | 6.5   | 5.7   | 1.6   | 0.4   | 0.0   | 0.0   |
|                                                                                  | PrecStd     | 13.3   | 1.8   | 1.9   | 1.6   | 3.9   | 4.6   | 7.0   | 7.9   | 7.3   | 4.6   | 3.5   | 2.4   | 1.1   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 77.2  | 82.3  | 84.7  | 89.0  | 92.6  | 94.0  | 95.3  | 96.2  | 94.3  | 91.3  | 86.0  | 79.8  |
|                                                                                  |             | MCWB   | 68.7  | 72.4  | 74.8  | 77.2  | 79.8  | 80.9  | 81.3  | 79.3  | 79.1  | 75.0  | 73.0  | 70.4  |
|                                                                                  | 2%          | DB     | 74.1  | 78.2  | 81.5  | 86.7  | 90.7  | 92.2  | 93.3  | 93.9  | 92.3  | 89.1  | 83.7  | 76.8  |
|                                                                                  |             | MCWB   | 66.4  | 70.5  | 73.8  | 77.1  | 79.4  | 81.0  | 81.0  | 80.2  | 78.8  | 74.8  | 71.7  | 68.4  |
|                                                                                  | 5%          | DB     | 71.6  | 75.3  | 78.8  | 84.9  | 89.0  | 90.7  | 91.7  | 92.1  | 90.8  | 87.3  | 81.7  | 74.3  |
|                                                                                  |             | MCWB   | 64.8  | 69.2  | 72.4  | 76.4  | 78.9  | 80.9  | 81.1  | 80.3  | 78.6  | 74.3  | 70.6  | 66.3  |
|                                                                                  | 10%         | DB     | 69.1  | 72.9  | 76.3  | 83.0  | 87.4  | 89.2  | 90.2  | 90.4  | 89.1  | 85.5  | 79.4  | 71.8  |
|                                                                                  |             | MCWB   | 63.8  | 67.9  | 71.3  | 75.7  | 78.5  | 80.7  | 80.9  | 80.3  | 77.9  | 73.8  | 69.9  | 64.5  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 70.7  | 73.7  | 76.6  | 80.1  | 83.8  | 84.1  | 84.3  | 83.9  | 82.5  | 79.5  | 77.2  | 74.6  |
|                                                                                  |             | MCDB   | 74.8  | 80.2  | 81.5  | 84.1  | 87.4  | 88.5  | 89.9  | 88.9  | 87.9  | 85.3  | 81.2  | 77.6  |
|                                                                                  | 2%          | WB     | 68.7  | 71.9  | 74.9  | 78.9  | 82.0  | 83.1  | 83.3  | 82.7  | 81.5  | 78.3  | 76.0  | 72.0  |
|                                                                                  |             | MCDB   | 72.3  | 76.5  | 79.9  | 83.8  | 87.1  | 87.8  | 88.8  | 88.3  | 87.2  | 84.2  | 79.9  | 74.9  |
|                                                                                  | 5%          | WB     | 67.0  | 70.5  | 73.6  | 77.9  | 80.9  | 82.4  | 82.6  | 81.9  | 80.7  | 77.4  | 74.8  | 69.3  |
|                                                                                  |             | MCDB   | 70.2  | 74.2  | 77.7  | 82.8  | 86.1  | 87.4  | 88.1  | 87.8  | 86.8  | 83.0  | 78.7  | 72.0  |
|                                                                                  | 10%         | WB     | 65.3  | 68.7  | 72.3  | 76.9  | 80.0  | 81.8  | 81.9  | 81.3  | 80.0  | 76.6  | 73.3  | 66.3  |
|                                                                                  |             | MCDB   | 68.4  | 72.2  | 75.7  | 81.8  | 85.2  | 86.7  | 87.4  | 87.2  | 85.9  | 82.0  | 77.6  | 69.9  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 10.1  | 9.6   | 9.4   | 9.6   | 10.0  | 9.1   | 9.4   | 10.5  | 11.0  | 11.6  | 12.1  | 11.6  |
|                                                                                  |             | MCDBR  | 14.2  | 13.0  | 12.4  | 11.3  | 11.9  | 11.1  | 11.8  | 13.3  | 13.4  | 13.6  | 14.3  | 14.8  |
|                                                                                  |             | MCWBR  | 7.7   | 6.5   | 6.2   | 4.6   | 4.0   | 3.3   | 3.2   | 3.9   | 3.9   | 4.5   | 5.8   | 7.5   |
|                                                                                  | 5% WB       | MCDBR  | 11.0  | 12.1  | 11.6  | 9.9   | 9.9   | 9.3   | 10.1  | 11.5  | 11.6  | 11.5  | 10.9  | 11.2  |
|                                                                                  |             | MCWBR  | 7.1   | 7.4   | 7.0   | 5.0   | 4.1   | 3.7   | 3.6   | 4.3   | 4.0   | 4.8   | 6.0   | 7.9   |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        |        | 0.621 | 0.656 | 0.812 | 0.768 | 0.572 | 0.544 | 0.522 | 0.586 | 0.636 | 0.733 | 0.613 | 0.633 |
|                                                                                  | taud        |        | 1.694 | 1.660 | 1.422 | 1.537 | 1.992 | 2.086 | 2.140 | 1.941 | 1.792 | 1.569 | 1.761 | 1.693 |
|                                                                                  | Ebn at Noon |        | 205   | 209   | 185   | 198   | 239   | 244   | 249   | 234   | 220   | 189   | 206   | 195   |
|                                                                                  | Edh at Noon |        | 70    | 77    | 102   | 92    | 58    | 52    | 49    | 60    | 69    | 83    | 65    | 68    |
| All-Sky Solar<br>Radiation                                                       | RadAvg      |        | 732   | 823   | 902   | 1178  | 1428  | 1409  | 1493  | 1501  | 1457  | 1314  | 1062  | 846   |
|                                                                                  | RadStd      |        | 175   | 148   | 134   | 139   | 130   | 192   | 175   | 87    | 89    | 90    | 149   | 146   |

## Historical Trends

|                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                        |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only           | N/A   | N/A     | N/A    | +1.08   | +0.47 | +0.36 | N/A         | N/A   | N/A   | N/A   |
| Regional (6 neighbors) | N/A   | N/A     | N/A    | +0.72   | N/A   | N/A   | N/A         | N/A   | N/A   | +112  |

Nomenclature: See separate page